# Violent Conflict and the Strength of Civil Society

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**Abstract:**
Leading theories of institutional and economic development emphasize the role of informal norms and the strength of civil society. Though informal norms and civil society are usually thought to change incrementally, exposure to violent conflict may shock these institutions. Well identified household level studies show that exposure to violent conflict may increase political participation, participation in social groups, involvement in community leadership, and contributions to public goods (Bauer et al. 2016). Evidence suggests that civil war may influence formal institutional quality at the country level, however, evidence that violence can change civil society or informal norms at the county level is sparse. In this study I apply the synthetic control method to model the impact of violent conflict on the strength of civil society at the country level. I focus on five countries in which existing micro-level evidence suggests that exposure to violence changes to behavior or informal norms: Sierra Leone, Burundi, Uganda, Nepal and Liberia. Results of the synthetic control analysis suggests that civil war is associated with stronger civil society in at least some contexts.

**Keywords:** Institutional Change, Violent Conflict, Informal Norms, Civil Society

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## I. Introduction

Institutions such as informal norms and the strength of civil society are important determinants of economic development. Violent conflict can shock the status quo leading to institutional changes; however, country level empirical evidence on the relationship between violent conflict and civil society is sparce. In this study I investigate whether violent conflict corresponds to changes to the strength of civil society at the country level.

Informal norms, including the strength civil society, are central to several leading theories of economic development. North (1990) describes the role of informal constraints on behavior that support formal institutional constraints. Informal constraints facilitate the protection of private property (Williamson and Kerekes 2011) and may substitute for formal rules when such rules are weak (Williamson and Mathers 2011). The strength of civil society plays an integral role in Fukuyama's (1996) theory of economic and political development. More recently, Acemoglu and Robinson (2019) describe the strength of civil society as one of two core aspects of institutions responsible for development. Much of the empirical evidence for the relationship between economic performance and institutional quality comes from studies that measure both formal institutions (Acemoglu and Johnson 2005; Knack and Keefer 1995; Acemoglu, Johnson, and Robinson 2001) and informal norms (Williamson 2009; Knack and Keefer 1997) at the country level.

Violent conflicts, such as civil wars, are large shocks to political equilibria, norms and even preferences, making these events a unique opportunity to study institutional change. Blattman and Miguel (2010) note the long run effect of violent conflict on economic development depends on the degree to which conflict influences institutional quality. Empirical



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evidence from cross-country studies supports the notion that violent conflict affects formal institutions. The risk of civil conflict may weaken constraints on the executive (Aguirre 2016) and weaken democracy and civil liberties (Chen et al., 2008). Brice (2018) finds mixed evidence of the effect of war on civil liberties, democracy and economic institutions. Higgs (1987) describes how crisis and specifically wars increased the size and scope of the United States government; likewise cross-country evidence suggests that civil wars increase the scope of government regulation (O'Reilly and Powell, 2015).

Evidence that violent conflict changes informal norms tends to come from individual and household level studies. In a meta study, Bauer et al. (2016) find that exposure to violence is associated with a variety of pro-social behaviors. Exposure to violence has been shown to increase social group participation, participation in community leadership, pro-social behavior in trust games, among other behaviors.[2] These studies have convincing identification strategies but have limited external validity because they can only identify localized effects on those exposed, not whether these household level effects result in changes at the country level.

In this study, we test if experiencing a violent conflict in the form of civil war is associated with a stronger civil society at the country level. Specifically, I focus on five countries for which there is robust household level evidence of changes in informal norms: Burundi, Liberia, Nepal Sierra Leone, and Uganda. The heterogenous nature of violent conflict poses a challenge to modeling its effect on the development process. Though most cross-country studies treat violent conflict as homogeneous, an exception is Bove, Elia, and Smith (2017) who model

[2] Section IV discusses these studies in more detail.



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the effect of civil war on GDP per capita using the common correlated effects estimator and the synthetic control method. Using a similar methodology, O'Reilly (2021) finds that civil war can influence the quality of institutional protections of property rights. In this study I follow O'Reilly (2021) to construct a counterfactual for the strength of civil society following civil war. The results show that for at least some countries civil war is associated with an increase in an index measuring the strength of civil society, though the effect is heterogeneous across countries.

The next section discusses various theories of how changes to behavior at the individual or household level may map into broader changes to informal norms or civil society. Section III discusses some of the challenges of modeling institutional change and describes the synthetic control method which is used to construct a counterfactual. The results of five case studies are presented in Section IV. Section V interprets the results and concludes.

## II. Theories of Violent Conflict and Institutional Change

Violent conflict may lead to institutional change through various channels. Economists usually take preferences as given, but a large shock (such as exposure to violence) may be such a formative experience it may change the underlying beliefs and preferences of individuals. Indeed, a literature in psychology studies the long-lasting effect of traumatic events on individuals. Psychological research describes how traumatic experiences can lead to anxiety, depression or more severe psychiatric problems, though Tedeschi and Calhoun (2004) discuss the possibility of more positive outcomes in what they refer to as "post-traumatic growth." Blattman (2009) and Bellows and Miguel (2009; 2006), among others in the political science literature, find that traumatic experiences can lead to greater political participation. If the



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proportion of those affected is large, these individual change in beliefs and preferences may map into boarder changes to the strength of civil society or institutions.

Rather than focusing on how violence directly changes individual preferences, exposure to violence may influence social preferences. One strand of evolutionary theory suggests that intergroup competition will lead to selection for either individual psychological traits or social norms that promote the success of the group, see Wilson (2012) for a discussion of the literature. Psychological responses or social norms that favor the in-group over outsiders will therefore be selected. Alternatively, individual selection may also lead to norms that favor the in-group. Through either channel, to the extent that exposure to violence increases the salience of intergroup competition, it may induce changes in social preferences or norms. See Bauer et al. (2016) for a more in-depth discussion of this theoretical mechanism and a review of the empirical literature supporting the theory that exposure to violence increases pro-social behavior, in particular pro-social behavior toward the in-group.

Finally, the neo-classical framework proposes that institutional change will occur due to changes in relative prices or incentives (North 1990). Organizations and individuals choose to operate (and maximize) within a given institutional framework or to maximize by attempting to change the institutional framework. Within the neoclassical framework war or violent conflict can change institutions if the conflict changes relative prices, which may include factor prices, changes to technology, the cost of information or the cost of bargaining (North, 1990: 83-84).

Applications of the neo-classical framework in the context of violent conflict include Olson's (2000) discussion of crises and interest groups and Higgs (1987) case study of institutional change in the United Sates. Others have studied how types of institutions may



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substitute for one another. Informal institutions tend to fill the gap in cases where formal institutions are weak or constrained (Leeson, 2007). Méon and Sekkat (2015) find evidence of substitution between informal social trust and formal legal institutions in determining investment. Similarly, Williamson and Mathers (2011) suggest that culture and may substitute for formal economic institutions. Alternatively, Acemoglu and Robinson (2019) model institutional change as a saddle path (Acemoglu and Robinson 2017) where formal institutions of the state complement informal norms of civil society and a balance between the two – implicitly at the country level – facilitates the process of development. Violent internal conflict may weaken state capacity (Besley and Persson 2008), therefore within the Acemoglu and Robinson framework, violence may inhibit the development of civil society.

Violent conflict may strengthen civil society directly by changing individual or social preferences or it may weaken or strengthen civil society indirectly through its effect on formal institutions (either by substitution or complementarity with informal norms). Though there are various theoretical channels by which conflict may influence informal institutions, whether the shock of violence conflict is sufficient to change broad norms or informal institutions remains an empirical question. The existing empirical evidence is at the household level which reveals little about the effect at the country level.

## III. Constructing a Counterfactual

### a. Empirical models of institutional change

Modeling institutional change empirically involves several challenges. First, institutional quality is difficult to measure. Studies such as Williamson (2009) rely on the World Values Survey or other survey data which is collected in infrequent waves and therefore provides a static



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or limited time dimension. Another obstacle is that most informal institutions or social norms are self-reinforcing and rooted in the history or culture of a society. Therefore, most institutional change is incremental and modeling the process of institutional change requires a measure of institutional quality over a substantial period of time.

Most cross-county studies of institutional change have applied panel fixed effects models and focused on changes to formal political (Aguirre 2016) or economic institutions (Lawson, Murphy, and Powell 2018; Brice 2018). An alternative approach to studying institutional change is country level studies, for instance, applying the synthetic control method to study how mass migration effects institutional quality in Jordan (Nowrasteh, Forrester, and Blondin 2020) and Israel (Powell, Clark, and Nowrasteh 2017).

b. Empirical models of violent conflict

Empirical studies of the effect of violent conflict or civil war on development must account for heterogeneity. Despite the clear destruction from civil war, studies that apply standard panel methods have found conflicting results about impact of civil war on economic performance (Collier 1999; Polachek and Sevastianova 2012; Biswas et al. 2016). The conflicting results may be because standard panel methods pool cross-sections together to estimate a common partial effect for all countries, masking the heterogeneous effects of violent conflict.

Case studies can account for heterogeneity but often lack a formal quantitative model. The absence of a formal counterfactual limits the inference can be drawn from case studies alone. Studying the economic effect of terrorism in the Basque region of Spain, Abadie and



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Gardeazabal (2003) overcome this limitation by using the synthetic control method to construct a counterfactual path of economic development. Bove, Elia, and Smith (2017) model the heterogeneity of civil wars in a panel setting using the common correlated effects estimator in addition to conducting synthetic control analysis of each civil war. Their results highlight the heterogeneous effects of civil war on economic performance. At the intersection of the literature on the effect of civil war and the literature on institutional change, O'Reilly (2021) applies the synthetic control method to model the effect of civil war on institutional quality, more specifically institutions related to the rule of law, access to justice and the protection of private property.

c. The synthetic control method

We model institutional change using the synthetic control method. Like other statistical techniques, the synthetic control method is used to estimate a treatment effect, $\tau_1$, as the difference between the treated observations, $y_1$, and a set of weighted untreated comparator observations, $y_i$.

$$\tau_1 = y_1 - \sum_{i=2}^{N+1} w_i y_i$$

The character of institutions is determined by history, culture, and tacit knowledge (Boettke, Coyne, and Leeson 2008). Given the determinants of institutions, constructing a counterfactual path of institutions that places weights on all observed units, as occurs in regression analysis, would likely place weights on irrelevant observations. The synthetic control method avoids comparing dissimilar units in two ways. First, the researcher restricts the donor pool – the set of countries that are included in the analysis. Then a counterfactual is created as a weighted average



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of comparison units (countries in this case). The distance, $D$, between a vector of covariates, $X$, for the treated country and the synthetic control country is minimized by selecting weights, $W$. Countries that are dissimilar in terms of covariates may receive a zero weight, further mitigating comparisons to countries with different institutional contexts.

$$D = \left((X_1 - X_0 W) V (X_1 - X_0 W)\right)^{\frac{1}{2}}$$

Covariates, $X$, are chosen to predict pretreatment values of institutional quality and matrix $V$ is a set of covariate weights determined by the degree to which they predict pretreatment institutional quality. Following Abadie, Diamond, and Hainmueller (2015), among others, we use an iterative optimization procedure to minimize the root mean squared prediction error between the pre-treatment institutional quality of the treated and comparator observations.

Once a counterfactual is constructed, a treatment effect can be estimated as in equation 1. The statistical significance of the treatment effect can be analyzed through in-space-placebo tests in which the treatment effect is estimated for non-treated observations. In addition, an F-test is conducted to assess the joint significance of the treatment effect for all post-treatment periods.

d. Measuring civil war and civil society
    i. Civil war

As discussed in the previous section, the effect of violent conflict on economic development is heterogenous. Further, several well identified studies have already established that exposure to violent conflict can influence pro-social behavior at the sub-national level. Therefore, the objective of this study is not to estimate an average effect of violent conflict on civil society at



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the country level. Instead, the objective of this study is to answer the question, "can violent conflict in the form of civil war effect the strength of civil society at country level?"

Since we are uninterested in the average effect of violent conflict, we study cases in which violent conflict is likely to affect the strength of civil society. We only consider cases in which there exists household or community level evidence of an effect of exposure to violent conflict on behavior or informal norms. In a meta-study Bauer et al. (2016) considers 19 micro-level studies, across 10 different countries, of the effect of violent conflict on behavior or informal norms.[3] Synthetic control analysis requires a significant number of pre-treatment periods to fit a model.[4] Of the 10 countries considered in the Bauer et al. meta-study, Georgia, Tajikistan, Bosnia Herzegovina, Israel, and Indonesia are excluded from our analysis because the pre-treatment period occurs before sufficient data become available. The five countries that remain are the focus of the present study: Burundi, Liberia, Nepal, Sierra Leone and Uganda.

Civil wars often have ambiguous start dates which can pose challenges for identifying causal effects of violent conflict. The Armed Conflict Dataset produced as part of the Uppsala Conflict Data Program (UCDP) includes data on the violence threshold reached in a given year (25 or 1000 battle related deaths), as well as the year that a conflict reaches a cumulative intensity of 1000 battle related deaths. Bauer et al. (2017) and O'Reilly (2021) use the lower 25 battle related death threshold to define conflict onset. Because we are interested in whether exposure to violence can lead to broader changes in norms, we choose to use a higher violence threshold of the year that a war reaches a cumulative intensity of 1000 battle related deaths as the

[3] The count of 10 unique countries excludes the United States from (Shewfekt 2009) because this study investigates the sub-set of veterans returning to the United States from the Vietnam war. This count also excludes the study of 35 countries conducted by (Grosjean 2014).
[4] Following Bove et al. (2017) we use 15 pre-treatment years.



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start date. In all cases a conflict start date defined using the cumulative threshold is within three years of when an individual year reaches an annual threshold of 1000 battle related deaths. For completeness, the results of analysis using a 25-battle related death threshold to define conflict onset are presented in the appendix. Table 1 lists information about civil war intensity from the UCDP for each conflict.

The synthetic control method requires a donor pool of comparator countries that have not experienced treatment in the pre-treatment or the post-treatment period. In their application of the technique to civil war Bove, Elia, and Smith (2017) and O'Reilly (2021) study a period of 15 pre-treatment years and 10 post-treatment years between 1960 and 2010. We adopt their criteria of 15 pre-treatment years but extend the post treatment period to as many as 20 years for cases with available data. Due to data limitations the post-treatment period is limited to only 13 years for Burundi, nine for Nepal, and eight for Liberia.

[Table 1]

ii. Civil society

The strength of civil society is measured by two indexes from the Varieties of Democracy (V-Dem) dataset. The V-Dem dataset provides a long timeseries of institutional measures for a large cross-section of countries. We use two measures from the V-Dem dataset to measure the strength of civil society: the core civil society index and the civil society participation index.

The core civil society index seeks to answer, "How robust is civil society?" This index is constructed using a Bayesian factor model of the following three components: CSO participatory



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environment, CSO entry and exit, and CSO repression. The V-Dem codebook clarifies that "The core civil society index CCSI is designed to provide a measure of a robust civil society, understood as one that enjoys autonomy from the state and in which citizens freely and actively pursue their political and civic goals, however conceived." See Table 3 appendix for additional information about each of the components to the index and the V-Dem codebook for a full description (Coppedge et al. 2019). The civil society participation index measures concepts that are more closely related to the interplay between formal political institutions and civil society. Specifically, the index seeks to answer: "Are major civil society organizations (CSOs) routinely consulted by policymakers; how large is the involvement of people in CSOs; are women prevented from participating; and is legislative candidate nomination within party organization highly decentralized or made through party primaries?" The index is constructed from a Bayesian factor model of the following four components: candidate selection (national/local), CSO consultation, CSO participatory environment and CSO women's participation. Because using this index would be an imperfect test of our hypothesis, we present the results using the civil society participation index as a robustness check in the appendix.

Components or indicators from V-Dem are coded by country experts who tend to be academics or professionals who have knowledge of the subject area, most of whom are nationals of or live in the country they are coding. V-Dem strives for five experts to code each country-year observation and aggregates scores from difference coders with a Bayesian measurement model which also adjusts for potential bias; see Pemstein et al. (2021) for details of the measurement model.

[Table 3]



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### iii. Control variables

The weights used to construct the counterfactual synthetic control country are based on how similar those countries are to the treatment country in term of a set of covariates. The covariates are selected and weighted based on how well they predict the pretreatment values of the index of civil society.

Informal norms tend to be highly persistent and civil society is no exception. Therefore, pretreatment values of the civil society index are expected to be good predictors of civil society. Though some have used all pre-treatment values as covariates, for example Bove et al. (2017), Kaul et al. (2015) cautions that including all pre-treatment values can lead to bias because other relevant covariates may receive a zero weight. To balance this tradeoff we use, the first, the eight, and fifteenth pre-treatment values of the civil society index as covariates.

Other covariates are included to capture demographic, economic, political, and geographic characteristics of the country that may influence civil society. The log of GDP per capita is included as a covariate to account for the standard of living and the log of population is included to account for the size of the country (Feenstra, Inklaar, and Timmer 2015). Formal political institutions are intertwined with norms like civil society. As such, we include the measure of electoral democracy from V-Dem as a covariate.[5] Institutions are rooted in slow changing or fixed factors such as history and geography (Auer 2013; Boettke, Coyne, and Leeson 2008). To account for an aspect of history we included legal origin as a covariate (LaPorta, Lopez-de-Silanes, and Shleifer 2008). Two aspects of geography enter the analysis. First, the donor pool is restricted to include only countries in the same region as the treatment

[5] The V-Dem electoral democracy index seeks to answer, "To what extent is the ideal of electoral democracy in its fullest sense achieved?" See the V-Dem codebook for information of the subcomponents that make up this index (Coppedge et al. 2019).



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country. Second, the absolute distance from the equator is included as a covariate. Summary statistics for the covariates are presented in Table 3.

[Table 3]

## IV. Cases

We present five cases of civil war as empirical evidence on the relationship between violent conflict and informal norms of civil society. For each case we briefly summarize the existing literature on the link between violent conflict and informal norms or pro-social behavior which consists of household or community level evidence. Then we evaluate the possibility that civil war influences civil society at the country level by conducting synthetic control analysis.

For each case, we first evaluate the fit of the pre-treatment synthetic counterfactual. We discuss the countries and corresponding weights that make up the counterfactual and how similar the counterfactual is in terms of the covariate characteristics. The fit of the counterfactual is assessed by considering the root mean squared prediction error between the actual quality of civil society and quality of civil society of the synthetic counterfactual. Second, we discuss the treatment effect of the civil war on civil society by comparing the true post-treatment index of civil society to the counterfactual values of the index. Finally, we evaluate whether the measured effect is statistically significant using two methods. An in-space-placebo test is conducted to test if the effect is statistically significant for individual post-treatment years. We also test if the effect is significant for all post-treatment years in a joint significance test (F-test).

The average score on the civil society participation index in Sub-Saharan Africa is plotted in Figure 1 along with the score for the four Sub-Saharan countries under consideration. Figure 2 plots the average of the index of civil society participation for South Asia along with



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Nepal. Since 1960, scores have increased for each country under consideration and for both regional averages. However, to draw inference from these patterns of institutional change a more rigorous counterfactual is necessary.

[Figure 1]

[Table 4]

[Table 5]

a. Uganda
    i. Literature

Uganda experienced low level violence in the early 1970s involving coup-d'états and interventions from neighboring countries. The violence reached a cumulative intensity of 1000 battle related deaths in 1979, the same year that General Idi Amin was overthrown as leader of the country. In 1981 under the leadership of Yoweri Museveni the National Resistance Army launched an armed uprising against the government and ultimately captured the capital in 1986. High levels of violence continued throughout the 1980s until a peace agreement was reached in 1988. Though, violence continued under the infamous Joseph Kony and the Lord's Resistance Army (LRA). The LRA engaged in a campaign of terror and abductions throughout the 1990s leading to the displacement of more than 1 million people.

Household level studies of exposure to violence in Uganda tend to focus on the more recent violence perpetrated by the Lord's Resistance Army (LRA). De Luca and Verpoorten (2015a) find that district level violent conflict leads to a contemporaneous decrease in generalized trust and membership in associations. However, these aspects of social capital recover rapidly after the initial shock. Violence from the LRA is associated with both economic



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and psychological effects (Blattman and Annan 2010; Annan et al. 2011). Violence in Uganda also influenced norms and behavior. Blattman (2009) studies the behavior of youth who were abducted and have returned from displacement. After correcting for attrition in the sample, abductees are more likely to vote and to be community mobilizers.[6] Finally, De Luca and Verpoorten (2015b) show that violence can stimulate political discussion and local meeting attendance, though the fail to an effect on voting.

ii. Synthetic control results

We now turn to the synthetic control results to test if the violent conflict corresponds to stronger civil society at the country level as measured by the core civil society index. The synthetic counterfactual of Uganda consists of 58% Benin, 29% Malawi and 13% Tanzania. This counterfactual is slightly richer and more democratic than actual Uganda. The level of human capital is almost identical. Despite a seemingly good match in terms of covariates, the pre-treatment match with the core civil society index is relatively poor. Synthetic Uganda has a lower score on the index during the early pre-treatment period and a lower score than actual Uganda in the later pre-treatment period. This poor match is reflected by the high RMSPE of 0.12, no lower than the RMSPE between Uganda and the Sub-Saharan African average. Therefore, we are less confident in the inference that can be drawn in the case of Uganda.

As shown in Figure 2, the core civil society index increases immediately following treatment in Uganda but about a decade later the index also increases in synthetic Uganda. At the end of the period of study, the core civil society index almost identical in Uganda and synthetic

[6] A community mobilizer is defined as "elected members of the community who are responsible for organizing the community for daily or weekly meetings" (Blattman, 2009).



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Uganda. Given the small treatment effect and poor pretreatment fit of the model it is no surprise that neither the in-space-placebo test in Figure 3 or the joint F-test in Table 5 show a significant treatment effect. The analysis of Uganda does not find any evidence of a treatment effect from the violent conflict.

[Figure 2 and Figure 3]

a. Nepal
    i. Literature

Civil war in Nepal began 1996 with relatively low levels of violence. By 2001 the violence escalated with the formation of the rebel People's Liberation Army, the same year that the conflict reached the threshold of 1000 cumulative battle related deaths. Most of the violence took place between 2001 and 2005, reaching over 1000 battle related deaths every year between 2002 and 2005. In addition to casualties, more than 100,000 people were displaced due to the violence. The civil war was brought to a close with a peace agreement in 2006.

Community level evidence from Nepal shows a relationship between violence and informal norms. Gilligan, Pasquale, and Samii (2014) match communities that experienced violence to similar communities that did not experience violence to identify the effect of exposure on aspects of social cohesion. They find that violence intensity is associated with higher voter turnout and higher scores on a community group participation index. Further, communities effected by violence have higher rates of cooperation in a public goods game. Rather than focusing social trust, Juan and Pierskalla (2016) study the effect of violence on trust in the Nepalese government. They find that individuals from villages exposed to more violence



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tend to exhibit lower levels of trust in the national government. These studies show that exposure to violence influenced behavior and norms at the community level in Nepal.

ii. Synthetic control results

To assess if violence in Nepal is associated with similar effects on community participation at the country level, we construct a synthetic Nepal that consists of 47% Fiji, 41% Mongolia, and 12% Vietnam. This synthetic counterfactual is slightly richer, better educated and more democratic than actual Nepal. Despite these differences in development, the pre-treatment path of the civil society participation index tracks the true path closely, including a sharp increase in the late 1980s. With a RMSPE 0.09, the synthetic control for Nepal is a better fit than that of Uganda and a notable improvement over the Asian average with a RMSPE of 0.25.

Prior to the onset of high intensity conflict in 2001 the core civil society index declines from about 0.8 to less than 0.6, see Figure 4. Because the synthetic counterfactual does not decline, there is a negative treatment effect between 2001 and 2006. In 2006 the core civil society index increases to above its pre-treatment level and the synthetic counterfactual decreases, producing a positive treatment effect. The joint F-tests shows a significant difference between pre-treatment and post-treatment but placebo tests in Figure 5 show that only the first post-treatment year is statistically significant. The mixed effect Figure 4 makes it difficult to draw any strong conclusions. Robustness checks using the alternative start date of 1996 show similar results, though the civil society participation index show no treatment effect. Therefore, we do not find evidence that civil war has a consistent positive or negative effect on the strength of civil society in Nepal.



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[Figure 4 and Figure 5]

b. Burundi

i. Background and literature

The 1993 election in Burundi resulted in the first Hutu president Melchior Ndadaye, who, within months was assassinated. The assassination was the catalyst for thousands of genocidal killings and retaliatory killings which was ultimately the onset of the civil war. The civil war resulted in 300,000 dead and over 1 million displaced (Bundervoet et al., 2009). According to the Uppsala Conflict Data Program (UCDP), the onset of civil conflict occurred in 1994 and the fighting reached a cumulative intensity of 1000 battle related deaths in 1997 and was coded as a high intensity war involving more than 1000 annual battle related deaths stating in 2000. See Bundervoet et al. (2009) for a timeline of the fighting. The Arusha Peace Accords in 2000 began the process of peace, though violence continued until 2008.

Voors et al. (2012) pair data on conflict victimization in Burundi to the results from a series of field experiments conducted in 2009. They collect data on exposure to violence from surveys in 1998 and 2007 to construct community and household level measures of victimization. Individuals exposed to violence at the individual or community level tend to be more altruistic to their neighbors (Voors et al. 2012). Voors and Bulte (2014) attempt to unbundle institutional change in Burundi following the civil war with household level data collected in 1998 and 2007. They study three categories of institutions: local land tenure security, hostility toward formal institutions of the state, and the strength of social capital, trust and cooperation. Their results indicate that exposure to violence leads to more secure land tenure within the local informal land tenure system. Further, they find that social capital and trust are influenced by exposure to violence from outsiders. For those exposed to violence, cooperation is



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greater and the level of trust for community members is greater. Both results suggest an increased reliance on local informal institutions. Finally, Voors and Bulte (2014) find that exposure to violence leads to a worse perception of the quality of formal institutions of the state, including the army, the police, the local justice authority and the local political authority.

ii. Synthetic control results

In the baseline synthetic control model, counterfactual Burundi consists of 45% Malawi, 28% Gabon, and 26% Swaziland. This mix of comparison countries is quite similar to Burundi in terms of the covariate country characteristics as reported in Table 4. The synthetic counterfactual is more developed, with a higher income, higher level of human capital and a higher score on the electoral democracy index. As shown in Figure 6, the synthetic counterfactual tracks the civil society index closely despite a sharp change in the index during the pre-treatment period. The RMSPE of the synthetic control is an improvement relative to the RMSPE for the Sub-Saharan African average of 0.25.

The core civil society index does not change with the onset of war in 1996. The index increases slightly in the early 2000s only to decline in the latter half of the decade. The result that the measured strength of civil society in Burundi is lower than the synthetic counterfactual a decade after the onset of civil war. Placebo tests in Figure 7 show that in several years the decline is statistically significant from changes in civil society in other countries in the donor pool. However, the F-test fails to reject the null of joint significance. As a robustness check we conduct the same analysis using the civil society participation index – the results are presented in the appendix. The treatment effect with this alternative index is also negative.

In an additional robustness check a lower violence threshold of 25 battle related deaths is used to determine the onset of conflict and therefore treatment. Using the lower violence



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threshold, the start date of the conflict in Burundi 1991 instead of 1996. As shown in the appendix, using this earlier start date the treatment effect is still negative and is statistically significant according to both placebo tests and an F-test for joint significance. Taken together, we do not evidence that the civil war in Burundi is associated with an increase in the strength of civil society, instead the evidence suggests that civil war undermined the strength of civil society at the country level.

[Figure 6 and Figure 7]

c. Liberia

i. Background and literature

Liberia experienced two civil wars in the past 35 years. The first Liberian Civil war began in 1989 and resulted in the capture of the capital by the National Patriotic Front of Liberia in 1990, bringing Charles Taylor to power. Foreign interventions and attempts at peace agreements reduced the number of battle related deaths throughout the 1990s but instability and violence continued. The violence threshold during this earlier civil war never reached the threshold of 1000 battle related deaths as coded by UCDP. Further, if this earlier violence were to be considered "treatment" it would causes the synthetic control to underestimate the treatment effect of civil war. The second Liberian civil war began in 1999 with both forces entering from neighboring Ghana and uprisings within Liberia. Although violence was occurring as early as 1999, violence reached a threshold of 1000 cumulative battle related deaths in 2002 according to the UCDP dataset. The Accra Comprehensive Peace Agreement brought the war to a close in 2003.

Micro-level evidence of the effect of violence on norms and behavior in Liberia includes Hartman and Morse (2018) who study altruism toward refugees from Ivory Coast by Liberians.



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Their results indicate that Liberians that experienced violence during their civil war were more altruistic. Specifically, they hosted more refugees and a higher proportion of refugees different from themselves in terms of religion and ethnicity. (Fearon, Humphreys, and Weinstein 2009) also studies post-conflict altruism in Liberia but instead focuses on "community driven reconstruction" programs. They find that villages randomly treated with a community driven reconstruction program give more in a public goods game. Whether through exposure to violence or aid programs implemented after exposure to violence, the microeconomic evidence suggests that some Liberians are exhibit more altruistic behavior following the civil wars.

ii. Synthetic control results

The counterfactual synthetic Liberia is constructed from 49% Swaziland, 48% Tanzania and 2% Malawi. Synthetic Liberia is more democratic and richer that actual Liberia. The synthetic strength of civil society tracks the true path closely during the pre-treatment period with a RMSPE of 0.04 compared to the Sub-Saharan African average RMSPE of 0.8. As shown in Figure 8, soon after the onset of high intensity civil war in Liberia between 2002 and 2003, the core civil society index increases whereas the index is stable for the counterfactual. This produces a substantial positive treatment effect of about 0.3 for most post-treatment periods, which is equivalent to about one standard deviation the full sample. The in-space-placebo test shows that the increase in the index after treatment is unusual relative to other countries in the donor pool. The effect is statistically significant in six of the eight post-treatment years as shown in Figure 9. Further, a joint significance test of all post-treatment periods shows that the effect of treatment is significant in Liberia.



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Robustness checks using the civil society participation index find a similar positive treatment effect. The alternative civil society participation index yields similar results as shown in the appendix. The results of the synthetic control analysis indicate that high intensity civil war in Liberia is associated with an increase in the strength of civil society at the country level. If the lower violence threshold is used to select the start date, the conflict is defined to start in 1989 with the onset of the first Liberian civil war. The results using this earlier start date are presented in the appendix show a negative but insignificant treatment effect. We do find evidence of a positive treatment effect in Liberia, but either effect of civil war on the strength of civil society occurs only later – when higher intensity fighting was taking place – or the primary results are not robust.

[Figure 8 and Figure 9]

d. Sierra Leone

i. Literature

The catalyst for the civil war in Sierra Leone was the invasion of by the Revolutionary United Front (RUF) from Liberia in 1991. The onset of the civil war occurred in 1991 and fighting reached a cumulative intensity of 1000 battle related deaths in 1995, the threshold used by the UCDP dataset for a high intensity conflict. Fueled by contestable diamond wealth (Keen 2005) the civil war continued until 1999, killing 50,000 people and displacing close to 1 million (Bellows and Miguel, 2006).

Using nationally representative survey data from post-conflict Sierra Leone, Bellows and Miguel (2009) study the effect of exposure to violence on political participation and other aspects of social capital. They suggest that exposure to violence may influence individual beliefs



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and preferences or give rise to changes to social norms and institutions. Specifically, households that were victimized by conflict in Sierra Leone were more likely to vote and were more likely to attend community meetings (Bellows and Miguel 2006; 2009). Further, those that experienced more intense violence were more likely to join political groups and community groups as well as more likely to contribute to local public goods (Bellows and Miguel 2009).[7] Similarly, Koos (2018) finds that households exposed to conflict-related sexual violence are more likely to be members of local associations and more likely to contribute funds to local communal events.

A decade after the civil war in Sierra Leone (Bauer et al., 2014) conducted experimental studies in Georgia and Sierra Leone to determine if those exposed to conflict were more likely to exhibit egalitarian behavior toward members of the "in-group" versus "out-group." Results from the sharing game indicate that those exposed to the war made more egalitarian choices when dealing with a member of the "in-group" while there was no increase in egalitarianism for interactions with the "out-group." Cecchi et al. (2016) find similar in-group and out-group dynamics in a study of youth football players in Sierra Leone. Finally, Cilliers, Dube, and Siddiqi (2016) show that post-conflict reconciliation programs in Sierra Leone strengthened social capital but that it came at the expense of psychological health.

ii. Synthetic control results

Synthetic Sierra Leone consists of 29% Malawi, 29% Zambia, 23% Gabon and 19% Tanzania. This counterfactual matches Sierra Leone closely in terms of GDP per capita and the score on the electoral democracy index. The counterfactual has a larger population and more human capital.

[7] See González and Miguel (2015) for a comment on potential omitted variable bias.



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The close match in-terms of covariates yields a close match in terms of the pre-treatment civil society index with a RMSPE of 0.06, an improvement over 0.08 for the regional average.

As shown in Figure 10, the core civil society index does not increase with the onset of high intensity civil war in Sierra Leone in 1993 but does increase nearly a decade later. The strength of civil society increases to a level greater than that of the synthetic counterfactual ten to 15 years after the onset of civil war yielding a positive treatment effect. In-space-placebo tests in Figure 11 indicate that the positive treatment effect is significant several post-treatment years, including year well after the end of the conflict. Further, the joint significance test in Table 5 is highly significant.

Robustness checks using the civil society participation index show a similar treatment effect that is marginally statistically significant. Finally, if the lower violence threshold is used to select the start date, treatment is set to 1991. The results are similar to the primary specification, showing a positive treatment effect that is statistically significant as measured by the joint F-test in Table A4. Though the results of a placebo test are significant in only the first two post-treatment years, see Figure A6. Taken together, the results suggest that civil war is associated with a modest but significant increase in the strength of civil society. The analysis of Sierra Leone is the strongest evidence of a positive association between civil war the strength of civil society at the country level.

[Figure 10 and 11]

## V. Interpretation and Conclusion



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Violent conflict is associated with changes to household level behavior and informal norms. However, this study fills a gap in the literature by testing whether violent conflict in the form of civil war is associated with changes to the strength of civil society at the country level. The results indicate that for at least some countries, civil war is associated higher values on an index of the strength of civil society. We identify the effect by conducting synthetic control analysis on five cases of civil war in Burundi, Liberia, Nepal, Sierra Leone and Uganda. Using the synthetic control method, the treatment effect is estimated for each case individually, therefore accounting for the heterogeneity of civil war. We find some evidence that civil war is associated with a weakening civil society in Burundi and a strengthening of civil society in Libera. The most persuasive evidence that civil war is associated with stronger civil society at the country level is from in Sierra Leone.

Our results reemphasize the importance of the various micro-level studies linking violent conflict to changes in behavior and norms by demonstrating that these localized effects may indeed lead to broader changes in the character of institutions. Finally, several leading theories of suggest that the strength of civil society is a fundamental determinant of institutional and economic development. By contributing to the literature on the determinants of the strength of civil society this study has implications for institutional and economic development more generally.



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Tables and Figures:

Table 1: Episodes of Violent Conflict

| Country | Low Intensity (25) | High Intensity (1000) | Cumulative Intensity (1000) |
|---|---|---|---|
| Sierra Leone | 1991-2001 | 1995,1998,1999 | 1993 |
| Burundi | 1991, 1992, 1994-2006, 2008 | 1999-2002 | 1996 |
| Uganda | 1971, 1972, 1974, 1979-1992, 1994-2010 | 1981-1989, 1996, 2002, 2004 | 1979 |
| Nepal | 1996-2006 | 2002-2005 | 2001 |
| Liberia | 1980, 1989, 1990, 2000-2003 | 2003 | 2002 |

Note: Conflict dates are obtained from the Armed Conflict Dataset from the Uppsala Conflict Data Program.

Table 2: Varieties of Democracy Component Definitions

| Index and Component Name | Description |
|---|---|
| Core Civil Society Index: | |
| CSO participatory environment | Which of these best describes the involvement of people in civil society organizations (CSOs)? |
| CSO entry and exit | To what extent does the government achieve control over entry and exit by civil society organizations (CSOs) into public life? |
| CSO repression | Does the government attempt to repress civil society organizations (CSOs)? |
| Civil Society Participation Index: | |
| Candidate selection, National/local | How centralized is legislative candidate selection within the parties? |
| CSO consultation | Are major civil society organizations (CSOs) routinely consulted by policymakers on policies relevant to their members? |
| CSO participatory environment | Which of these best describes the involvement of people in civil society organizations (CSOs)? |
| CSO women's participation | Are women prevented from participating in civil society organizations (CSOs)? |

Note: Variable definitions are from the V-Dem codebook (Coppedge et al. 2019)



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Table 3: Summary Statistics

| Variable | Mean | SD | Max | Min | N |
|---|---|---|---|---|---|
| Civil Society Participation Index | 0.55 | 0.29 | 0.99 | 0.02 | 6700 |
| Core Civil Society Index | 0.56 | 0.31 | 0.98 | 0.01 | 6700 |
| Log GDP per capita | 8.52 | 1.24 | 12.41 | 4.96 | 6406 |
| Log Population | 2.14 | 1.63 | 7.20 | -2.21 | 6406 |
| Electoral Democracy | 0.43 | 0.29 | 0.94 | 0.01 | 6682 |
| Distance | 0.29 | 0.19 | 0.72 | 0.00 | 6703 |
| Human Capital | 2.01 | 0.71 | 3.70 | 1.01 | 6406 |

Note: The civil society participation index, the core civil society index and electoral democracy are from the V-Dem dataset (Coppedge et al. 2019). Data on log GDP per capita, log population and human capital are from the Penn World Tables f(Feenstra, Inklaar, and Timmer 2015). Legal origin is from (LaPorta, Lopez-de-Silanes, and Shleifer 2008) and distance measures the absolute distance from the equator.



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Table 4: Fit of the Synthetic Control – Civil Society Participation

| | Treated Covariates | Synthetic Covariates | Weights | Synthetic Country | Weight | RMSPE |
|---|---|---|---|---|---|---|
| Burundi | | | | | | |
| Civil Society (t-1) | 0.48 | 0.60 | 0.06 | Malawi | 0.45 | 0.08 |
| Civil Society (t-8) | 0.12 | 0.14 | 0.21 | Gabon | 0.28 | |
| Civil Society (t-15) | 0.09 | 0.14 | 0.14 | Swaziland | 0.26 | |
| Log GDP per capita | 6.67 | 8.00 | 0.02 | | | |
| Log Population | 1.66 | 0.87 | 0.01 | | | |
| Electoral Democracy | 0.18 | 0.21 | 0.51 | | | |
| Distance Equator | 0.04 | 0.15 | 0.01 | | | |
| Legal Origin | 1.00 | 0.28 | 0.04 | | | |
| Human Capital | 1.13 | 1.56 | 0.00 | | | |
| | | | | RMSPE SSA: | | 0.25 |
| Liberia | | | | | | |
| Civil Society (t-1) | 0.56 | 0.55 | 0.97 | Swaziland | 0.49 | 0.04 |
| Civil Society (t-8) | 0.51 | 0.55 | 0.00 | Tanzania | 0.48 | |
| Civil Society (t-15) | 0.49 | 0.49 | 0.00 | Malawi | 0.02 | |
| Log GDP per capita | 5.99 | 7.85 | 0.00 | | | |
| Log Population | 0.84 | 1.64 | 0.00 | | | |
| Electoral Democracy | 0.25 | 0.27 | 0.01 | | | |
| Distance Equator | 0.07 | 0.18 | 0.00 | | | |
| Legal Origin | 1.00 | 1.00 | 0.02 | | | |
| Human Capital | 1.50 | 1.56 | 0.00 | | | |
| | | | | RMSPE SSA: | | 0.08 |
| Nepal | | | | | | |
| Civil Society (t-1) | 0.60 | 0.79 | 0.00 | Fiji | 0.47 | 0.09 |
| Civil Society (t-8) | 0.81 | 0.75 | 0.69 | Mongolia | 0.41 | |
| Civil Society (t-15) | 0.43 | 0.43 | 0.06 | Vietnam | 0.12 | |
| Log GDP per capita | 7.02 | 8.15 | 0.02 | | | |
| Log Population | 3.02 | 0.70 | 0.05 | | | |
| Electoral Democracy | 0.32 | 0.47 | 0.00 | | | |
| Distance Equator | 0.32 | 0.32 | 0.09 | | | |
| Legal Origin | 1.00 | 0.47 | 0.01 | | | |
| Human Capital | 1.32 | 2.31 | 0.07 | | | |
| | | | | RMSPE SSA: | | 0.23 |

Note: Results are obtained from estimating a synthetic control model as described in section III.



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Table 4: Fit of the Synthetic Control – Civil Society Participation continued

| | Treated Covariates | Synthetic Covariates | Weights | Synthetic Country | Weight | RMSPE |
|---|---|---|---|---|---|---|
| Sierra Leone | | | | | | |
| Civil Society (t-1) | 0.68 | 0.61 | 0.00 | Malawi | 0.29 | 0.06 |
| Civil Society (t-8) | 0.26 | 0.27 | 0.38 | Zambia | 0.29 | |
| Civil Society (t-15) | 0.26 | 0.24 | 0.34 | Gabon | 0.23 | |
| Log GDP per capita | 7.25 | 7.65 | 0.02 | Tanzania | 0.19 | |
| Log Population | 1.25 | 1.71 | 0.03 | | | |
| Electoral Democracy | 0.19 | 0.23 | 0.01 | | | |
| Distance Equator | 0.09 | 0.10 | 0.19 | | | |
| Legal Origin | 1.00 | 0.77 | 0.02 | | | |
| Human Capital | 1.19 | 1.51 | 0.00 | | | |
| | | | | RMSPE SSA: | | 0.08 |
| Uganda | | | | | | |
| Civil Society (t-1) | 0.19 | 0.26 | 0.10 | Benin | 0.58 | 0.12 |
| Civil Society (t-8) | 0.24 | 0.32 | 0.01 | Malawi | 0.29 | |
| Civil Society (t-15) | 0.57 | 0.40 | 0.03 | Tanzania | 0.13 | |
| Log GDP per capita | 6.85 | 7.34 | 0.01 | | | |
| Log Population | 2.28 | 1.44 | 0.05 | | | |
| Electoral Democracy | 0.17 | 0.20 | 0.58 | | | |
| Distance Equator | 0.02 | 0.11 | 0.00 | | | |
| Legal Origin | 1.00 | 0.42 | 0.00 | | | |
| Human Capital | 1.18 | 1.19 | 0.22 | | | |
| | | | | RMSPE SSA: | | 0.12 |

Note: Results are obtained from estimating a synthetic control model as described in section III.

Table 5: F-Test for Significance - Core Civil Society Index

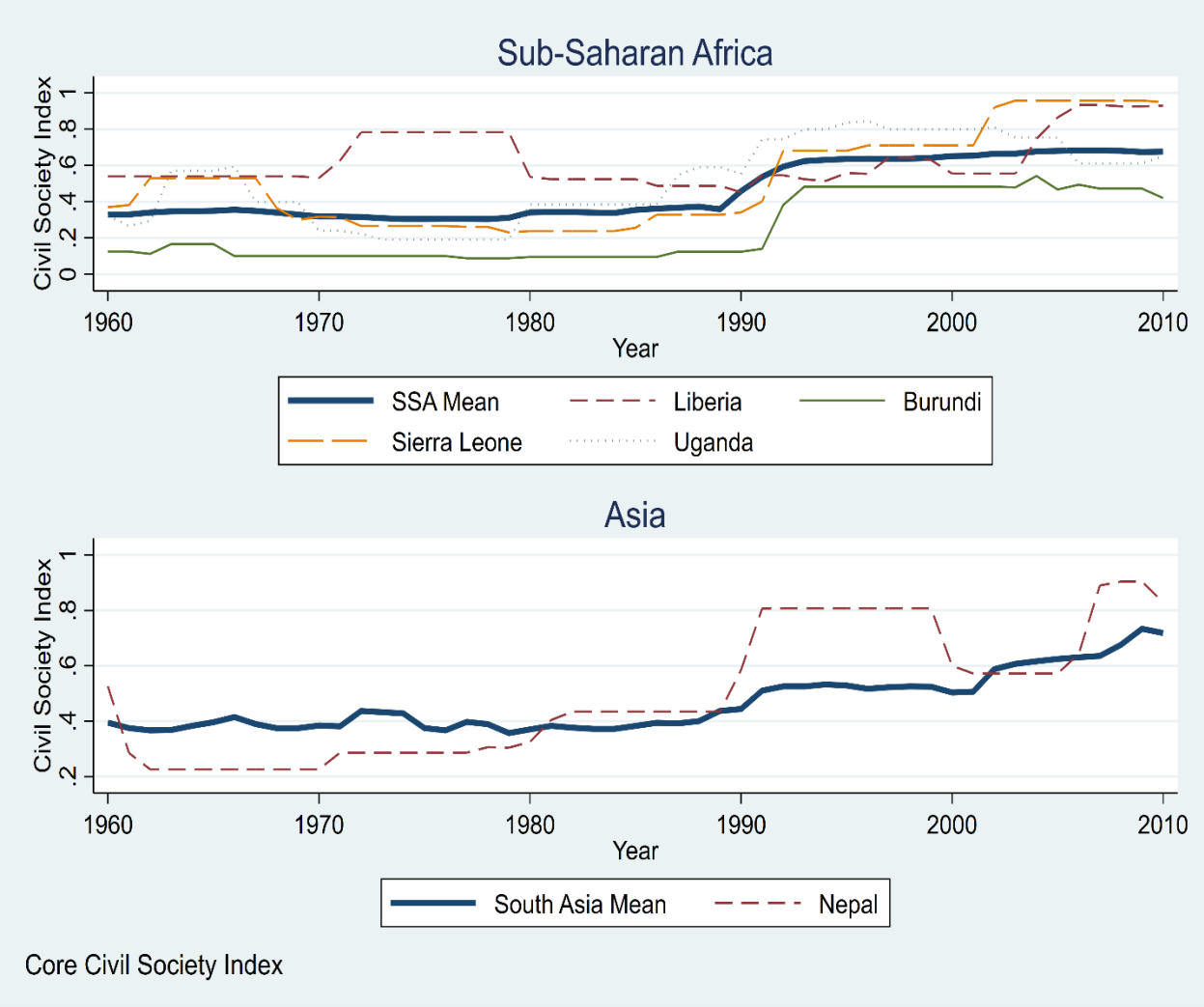
| | F-Statistic | P-value |
|---|---|---|
| Burundi | 1.41 | 0.263 |
| Liberia | 46.797 | 0.000 |
| Nepal | 10.293 | 0.000 |
| Sierra Leone | 4.49 | 0.003 |
| Uganda | 1.025 | 0.493 |

Note: The F-test is a joint significance test for all the post-treatment periods.



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Figure 1




Core Civil Society Index



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Figure 2

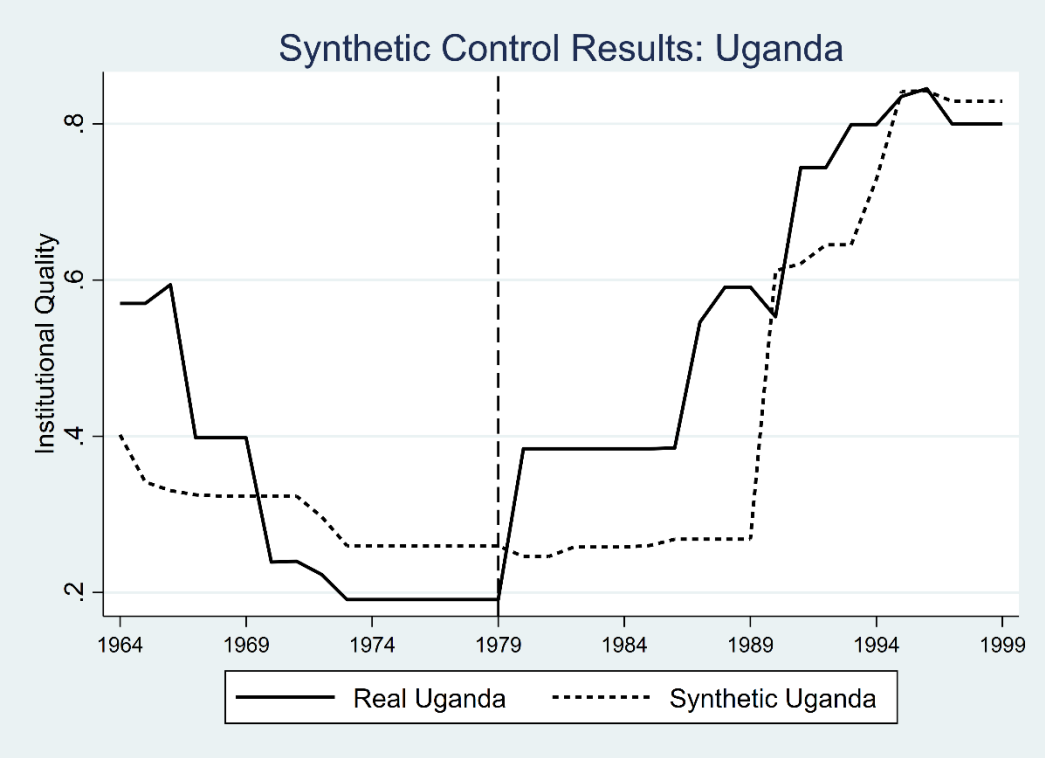
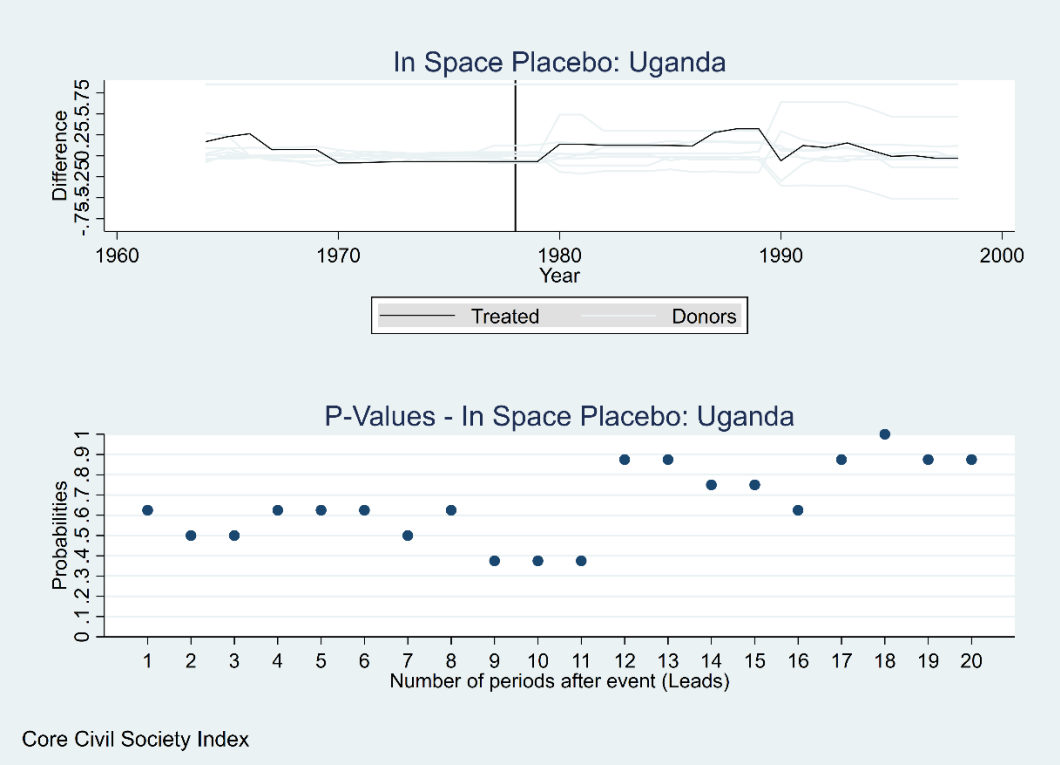



Figure 3






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Figure 4

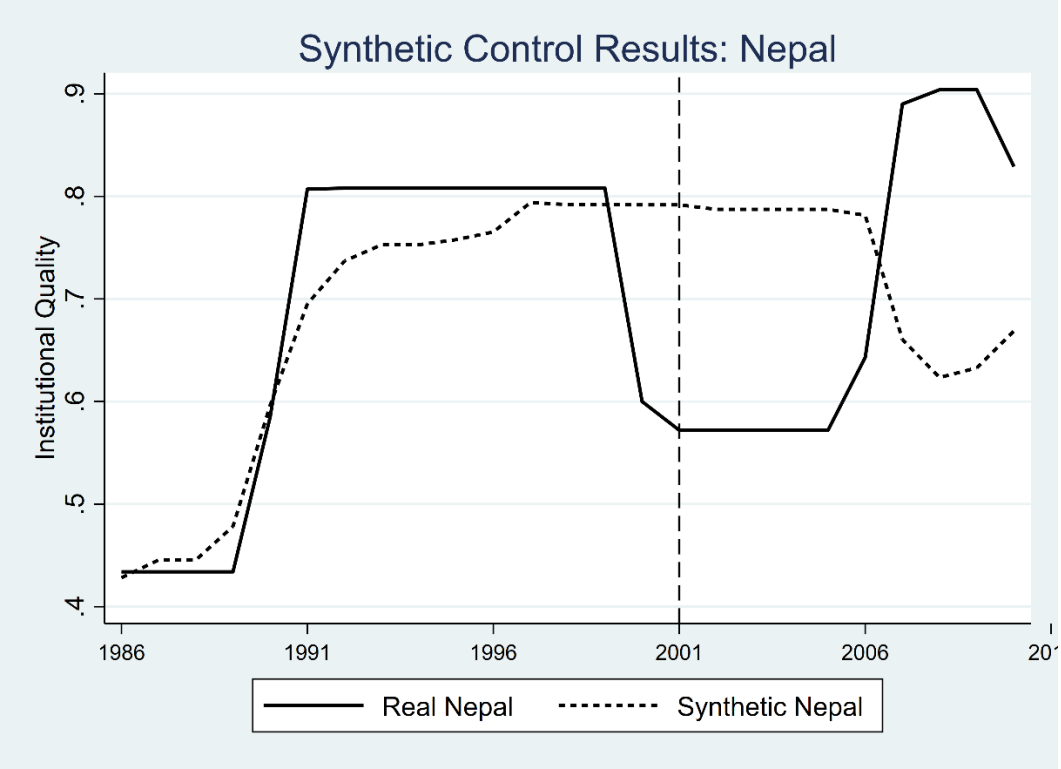
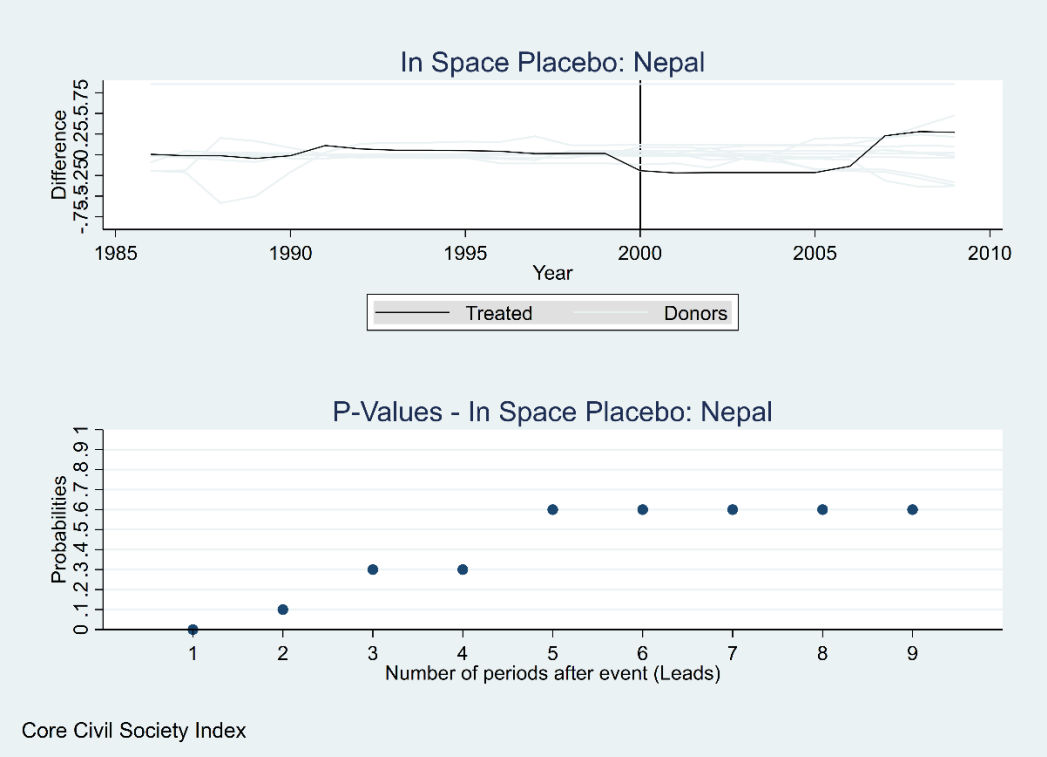



Figure 5






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Figure 6

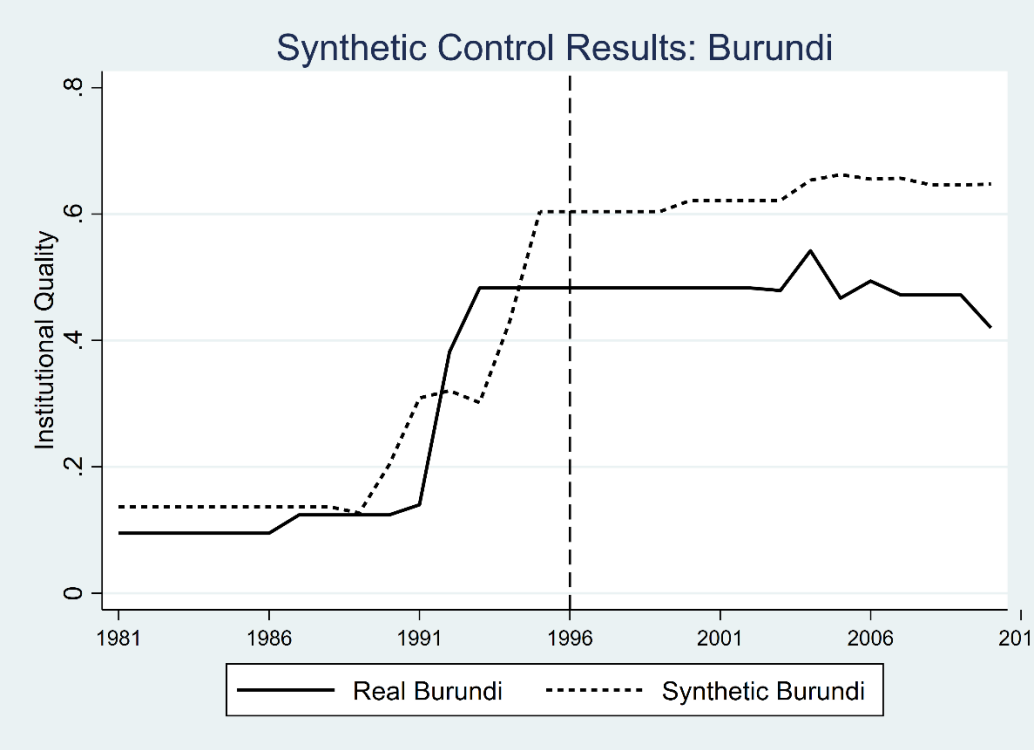
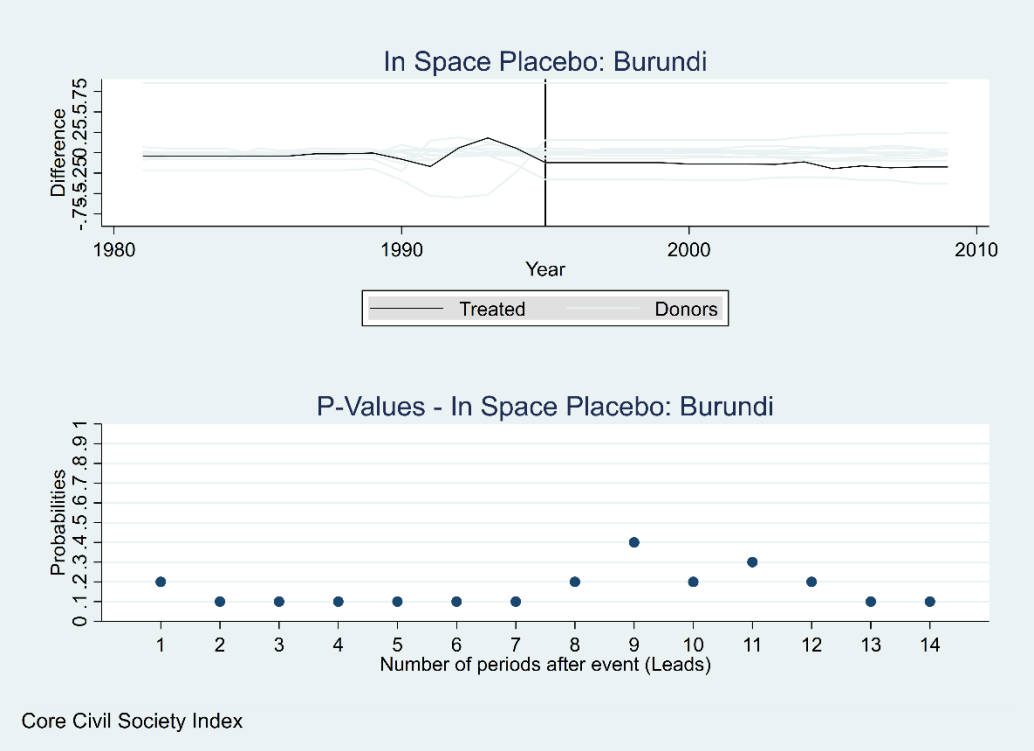



Figure 7






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Figure 8

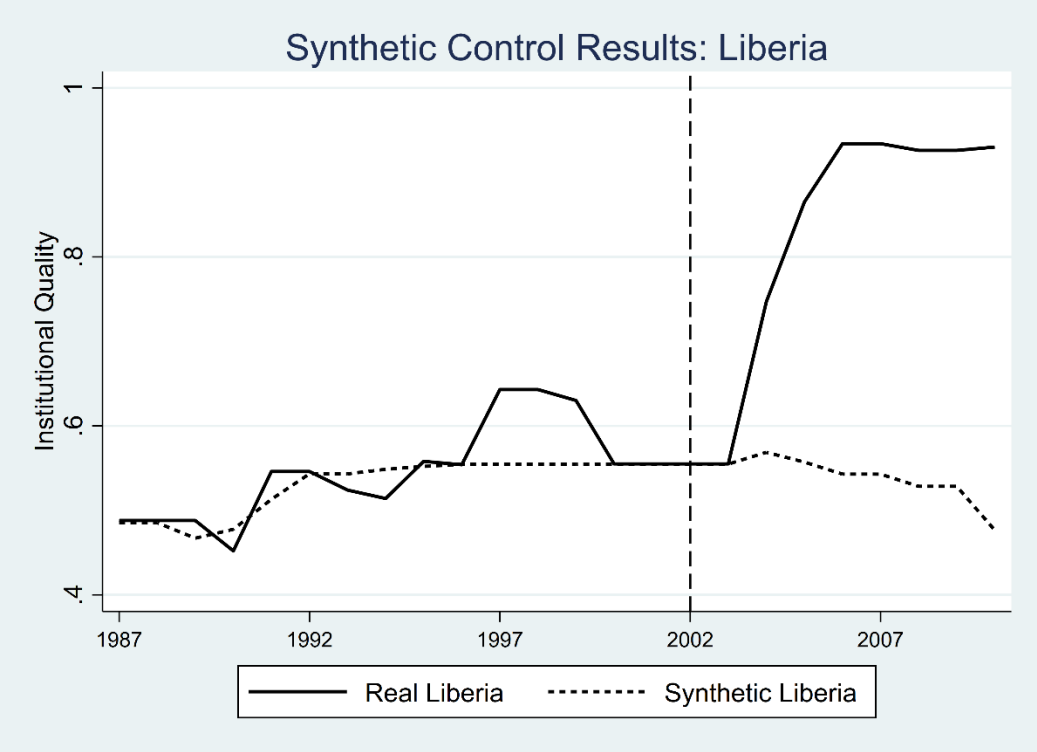
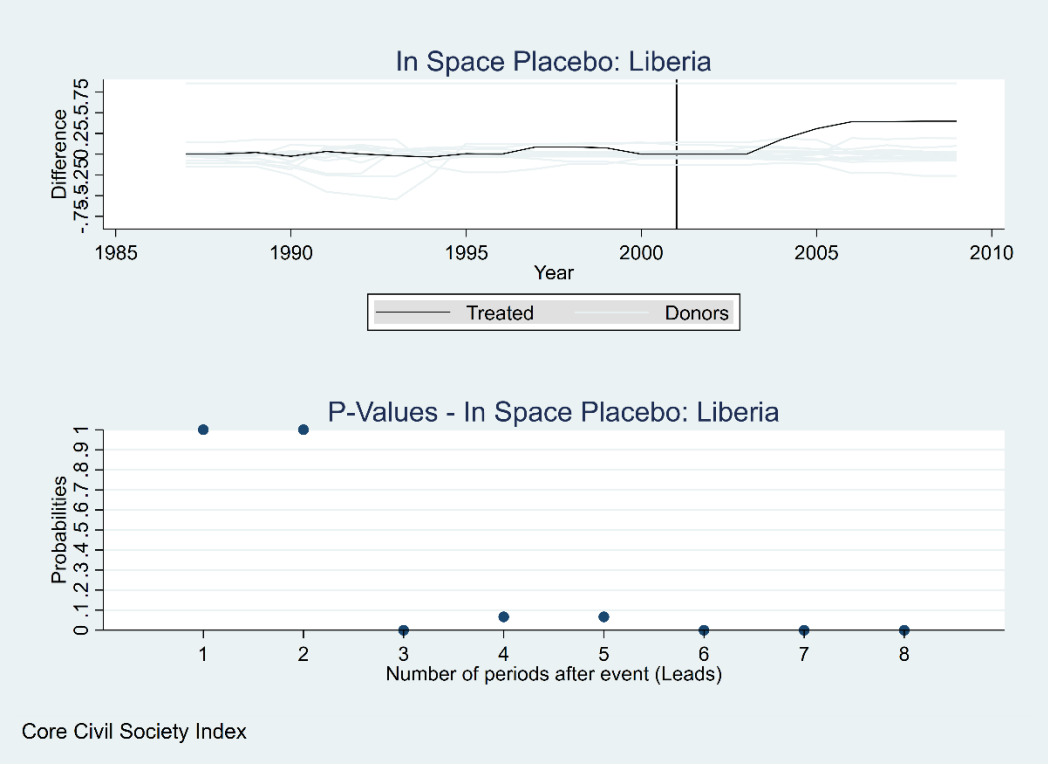



Figure 9






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Figure 10

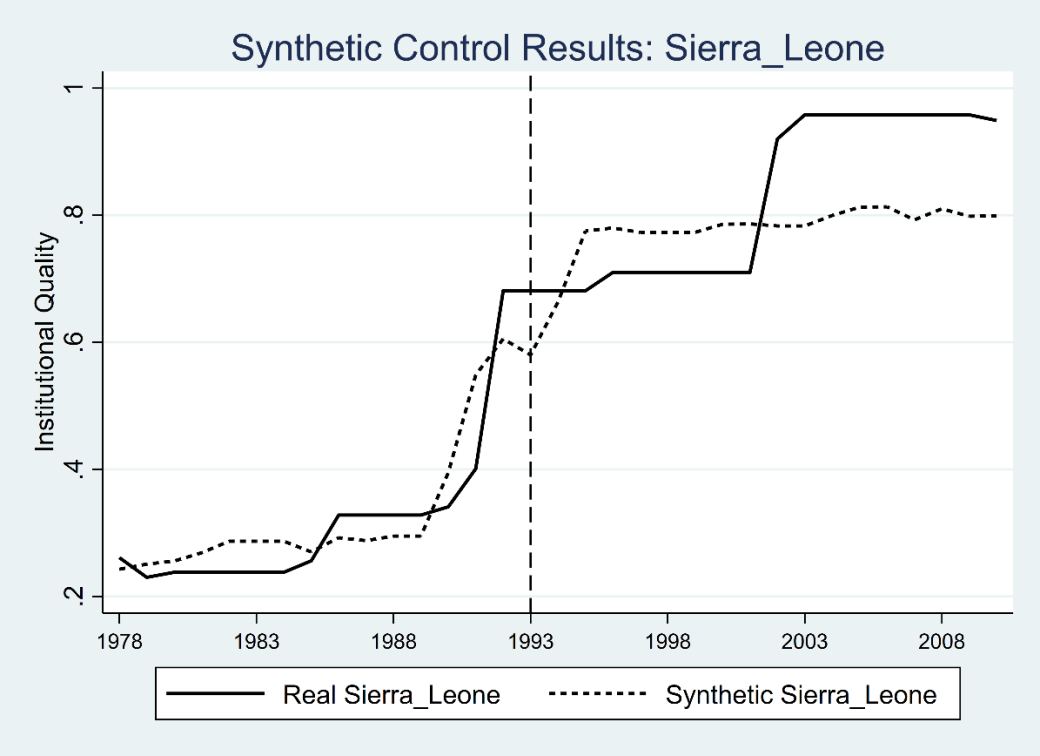
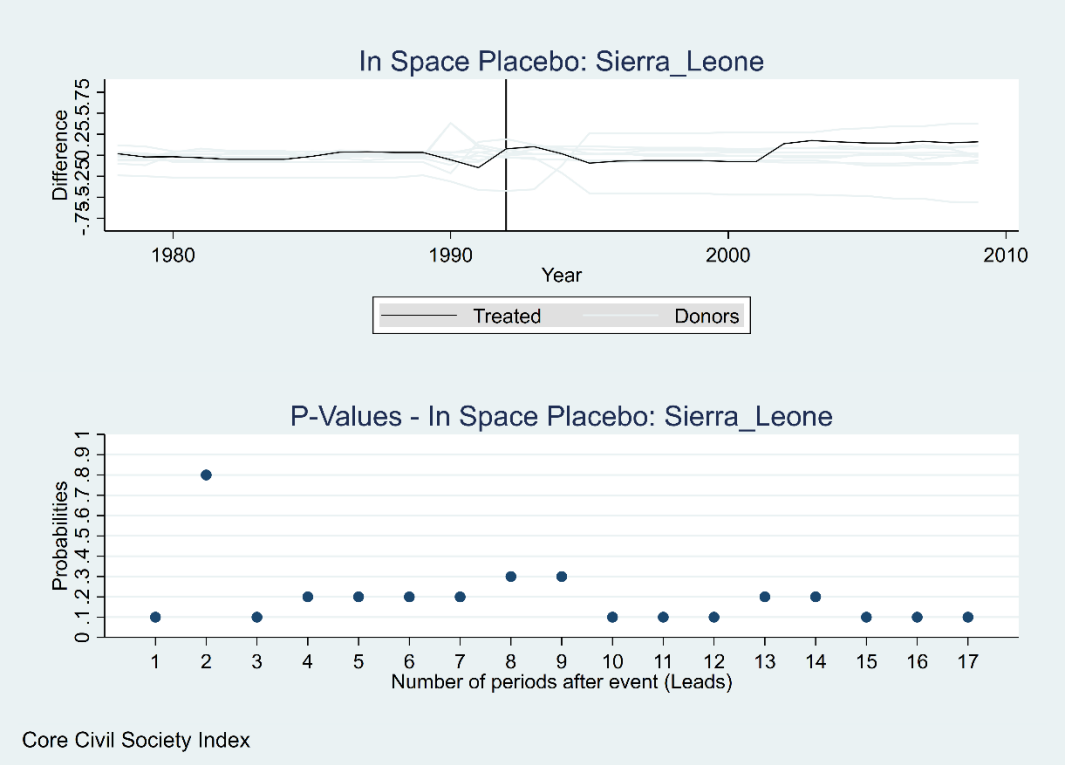



Figure 11



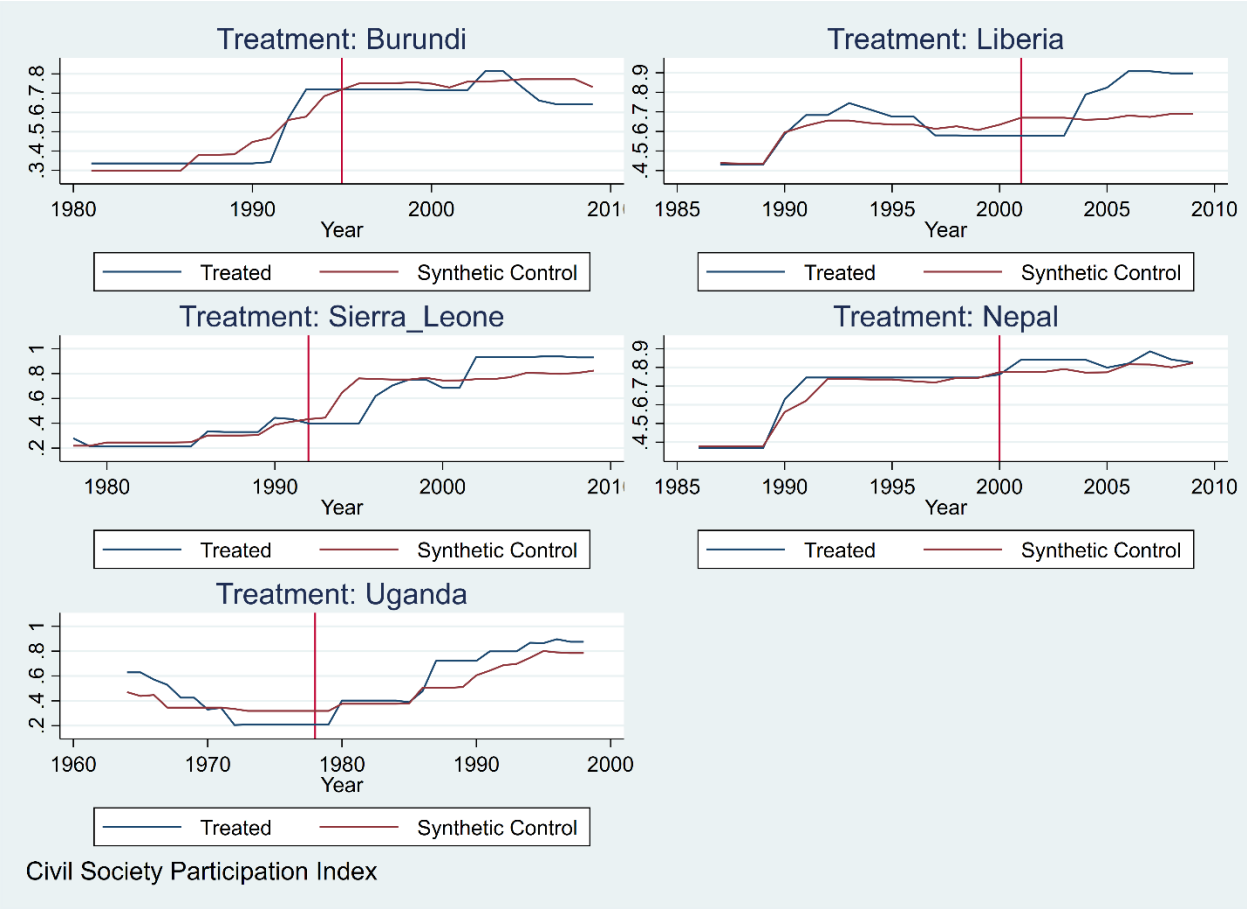



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Appendix:

Civil Society Participation Index – High Intensity

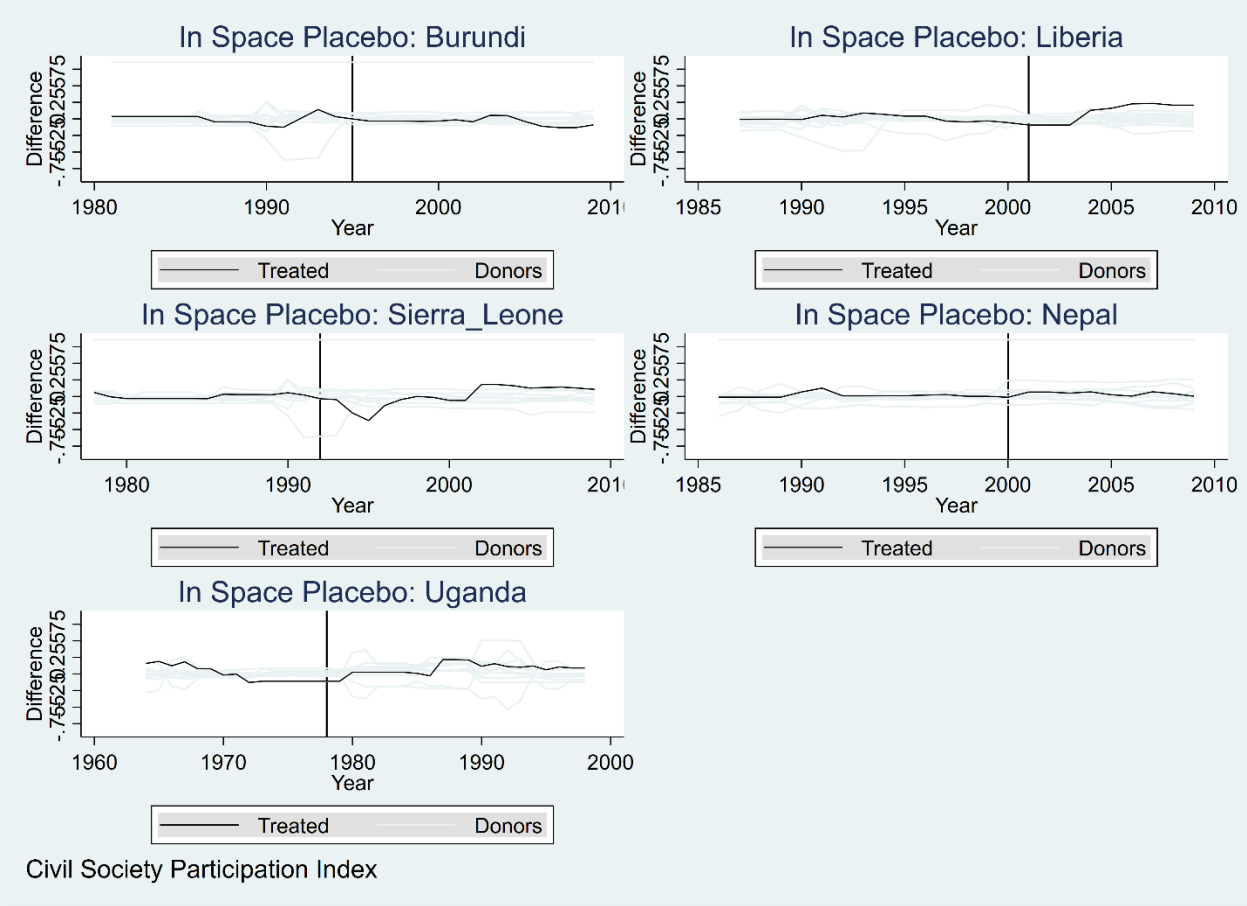
Figure A1: Synthetic Control Effects






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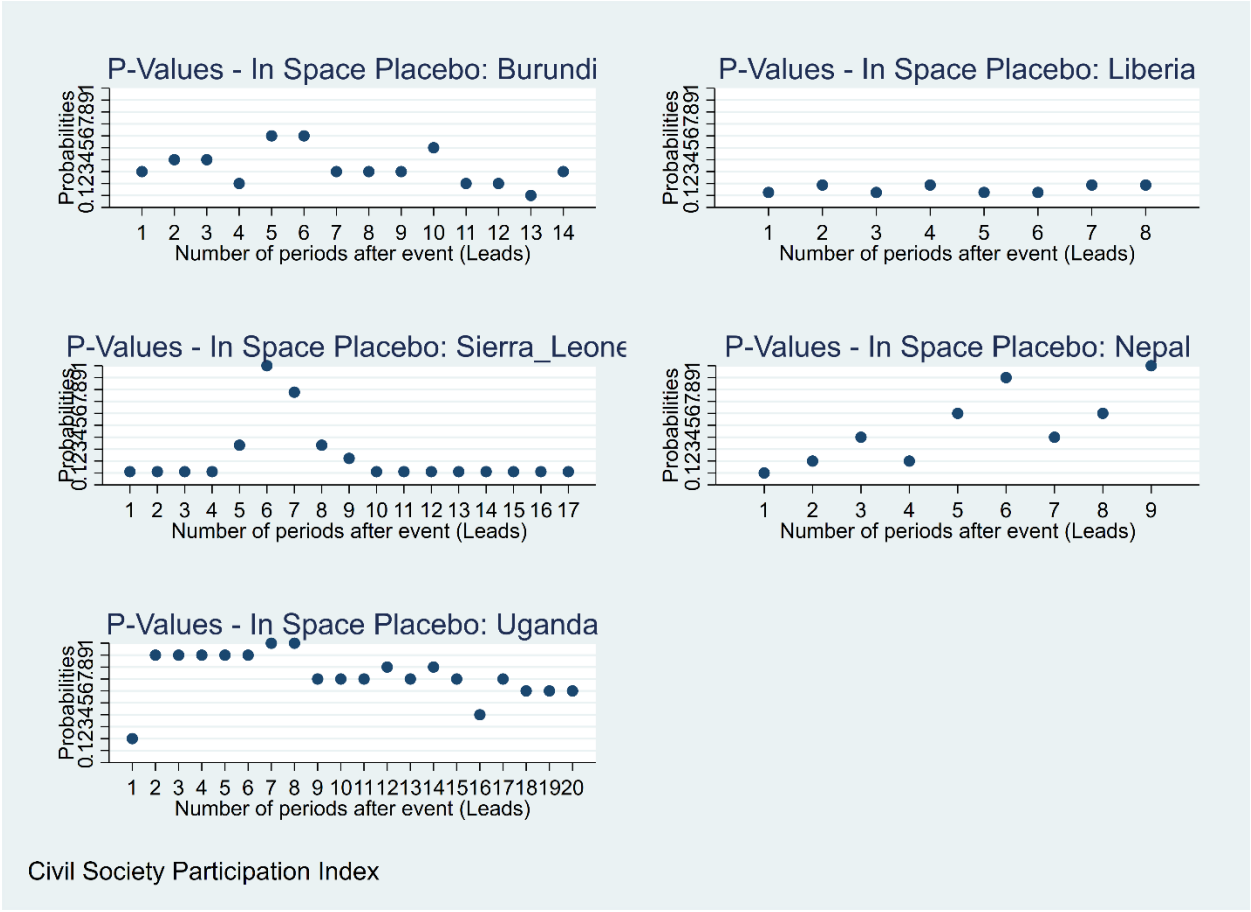
Figure A2: In Space Placebo Tests






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Figure A3: Placebo P-values






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Table A1: Fit of the Synthetic Control – Civil Society Participation Index (High Intensity)

| | Treated Covariates | Synthetic Covariates | Weights | Synthetic Country | Weight | RMSPE |
|---|---|---|---|---|---|---|
| Burundi | | | | | | |
| Civil Society (t-1) | 0.72 | 0.72 | 0.37 | Madagascar | 0.78 | 0.06 |
| Civil Society (t-8) | 0.34 | 0.38 | 0.16 | Malawi | 0.17 | |
| Civil Society (t-15) | 0.34 | 0.30 | 0.12 | Swaziland | 0.05 | |
| Log GDP per capita | 6.67 | 7.03 | 0.06 | Benin | 0.00 | |
| Log Population | 1.66 | 2.23 | 0.00 | | | |
| Electoral Democracy | 0.18 | 0.28 | 0.21 | | | |
| Distance Equator | 0.04 | 0.20 | 0.00 | | | |
| Legal Origin | 1.00 | 0.79 | 0.07 | | | |
| Human Capital | 0.72 | 0.72 | 0.37 | | | |
| Liberia | | | | | | |
| Civil Society (t-1) | 0.58 | 0.67 | 0.01 | Swaziland | 0.32 | 0.05 |
| Civil Society (t-8) | 0.71 | 0.64 | 0.02 | Togo | 0.28 | |
| Civil Society (t-15) | 0.43 | 0.44 | 0.11 | Benin | 0.25 | |
| Log GDP per capita | 5.99 | 7.71 | 0.00 | Gambia, The | 0.15 | |
| Log Population | 0.84 | 0.83 | 0.29 | | | |
| Electoral Democracy | 0.25 | 0.29 | 0.11 | | | |
| Distance Equator | 0.07 | 0.17 | 0.00 | | | |
| Legal Origin | 1.00 | 0.47 | 0.00 | | | |
| Human Capital | 1.50 | 1.51 | 0.46 | | | |
| Nepal | | | | | | |
| Civil Society (t-1) | 0.76 | 0.77 | 0.00 | Mongolia | 0.62 | 0.04 |
| Civil Society (t-8) | 0.75 | 0.74 | 0.49 | Maldives | 0.19 | |
| Civil Society (t-15) | 0.37 | 0.38 | 0.48 | Australia | 0.19 | |
| Log GDP per capita | 7.02 | 8.41 | 0.00 | | | |
| Log Population | 3.02 | 0.77 | 0.01 | | | |
| Electoral Democracy | 0.32 | 0.53 | 0.00 | | | |
| Distance Equator | 0.32 | 0.38 | 0.00 | | | |
| Legal Origin | 1.00 | 0.38 | 0.01 | | | |
| Human Capital | 1.32 | 2.51 | 0.00 | | | |

Note: Results are obtained from estimating a synthetic control model as described in section III.



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Table A1: Fit of the Synthetic Control –Civil Society Participation Index (High Intensity) continued

| | Treated Covariates | Synthetic Covariates | Weights | Synthetic Country | Weight | RMSPE |
|---|---|---|---|---|---|---|
| Sierra Leone | | | | | | |
| Civil Society (t-1) | 0.40 | 0.43 | 0.13 | Malawi | 0.49 | 0.04 |
| Civil Society (t-8) | 0.21 | 0.25 | 0.13 | Tanzania | 0.28 | |
| Civil Society (t-15) | 0.28 | 0.22 | 0.21 | Benin | 0.18 | |
| Log GDP per capita | 7.25 | 7.22 | 0.16 | Gabon | 0.03 | |
| Log Population | 1.25 | 2.12 | 0.00 | Swaziland | 0.02 | |
| Electoral Democracy | 0.19 | 0.22 | 0.24 | | | |
| Distance Equator | 0.09 | 0.12 | 0.09 | | | |
| Legal Origin | 1.00 | 0.79 | 0.03 | | | |
| Human Capital | 1.19 | 1.33 | 0.02 | | | |
| Uganda | | | | | | |
| Civil Society (t-1) | 0.21 | 0.32 | 0.07 | Tanzania | 0.64 | 0.12 |
| Civil Society (t-8) | 0.34 | 0.35 | 0.53 | Benin | 0.24 | |
| Civil Society (t-15) | 0.63 | 0.47 | 0.07 | Malawi | 0.12 | |
| Log GDP per capita | 6.85 | 7.43 | 0.03 | | | |
| Log Population | 2.28 | 2.16 | 0.00 | | | |
| Electoral Democracy | 0.17 | 0.24 | 0.19 | | | |
| Distance Equator | 0.02 | 0.09 | 0.00 | | | |
| Legal Origin | 1.00 | 0.76 | 0.07 | | | |
| Human Capital | 1.18 | 1.26 | 0.04 | | | |

Note: Results are obtained from estimating a synthetic control model as described in section III.

Table A2: F-Test for Significance

| | Core Civil Society Index | |
|---|---|---|
| | F-Statistic | P-value |
| Burundi | 0.85 | 0.622 |
| Liberia | 10.017 | 0.000 |
| Nepal | 0.812 | 0.623 |
| Sierra Leone | 18.027 | 0.000 |
| Uganda | 0.592 | 0.866 |

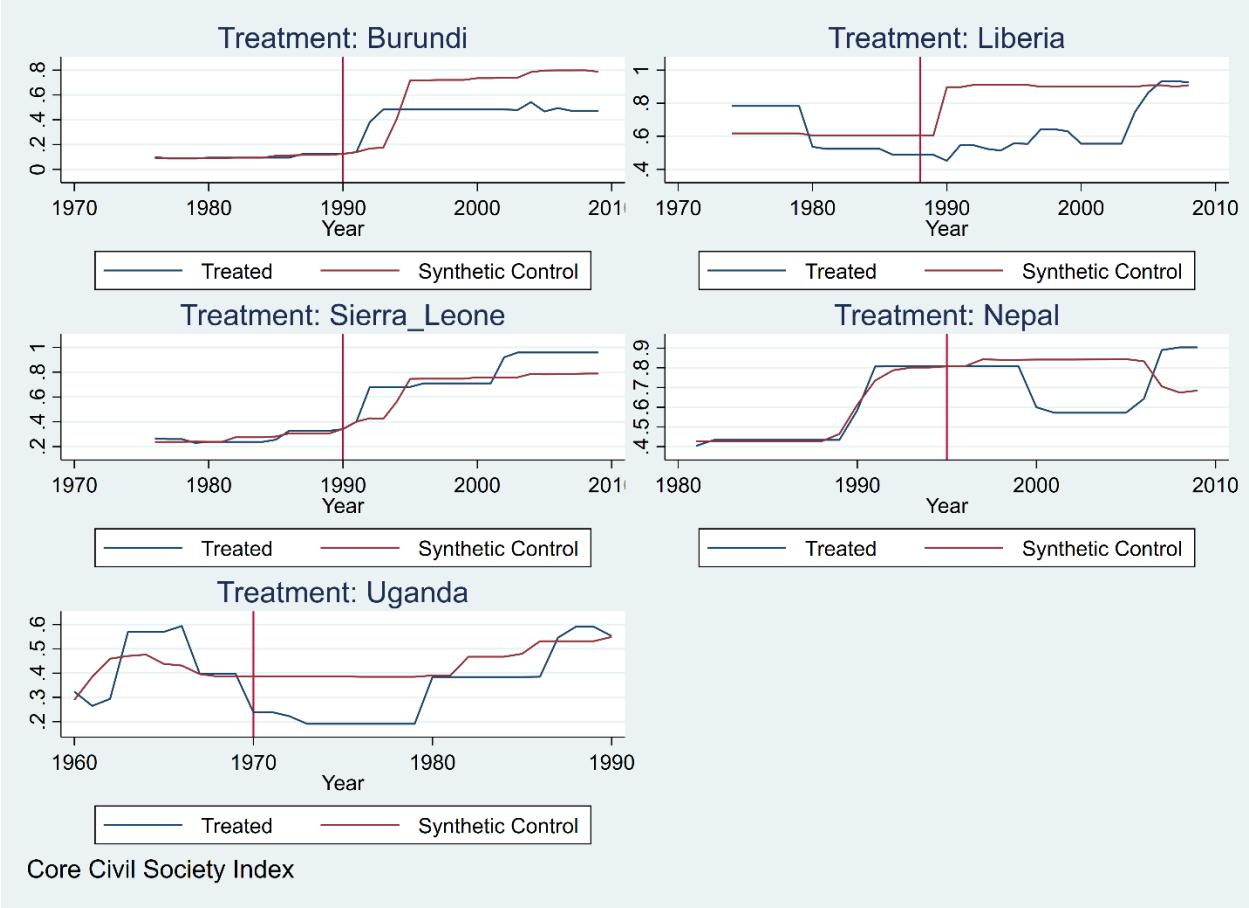
Note: The F-test is a joint significance test for all the post-treatment periods.



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Core Civil Society Index – Low Intensity

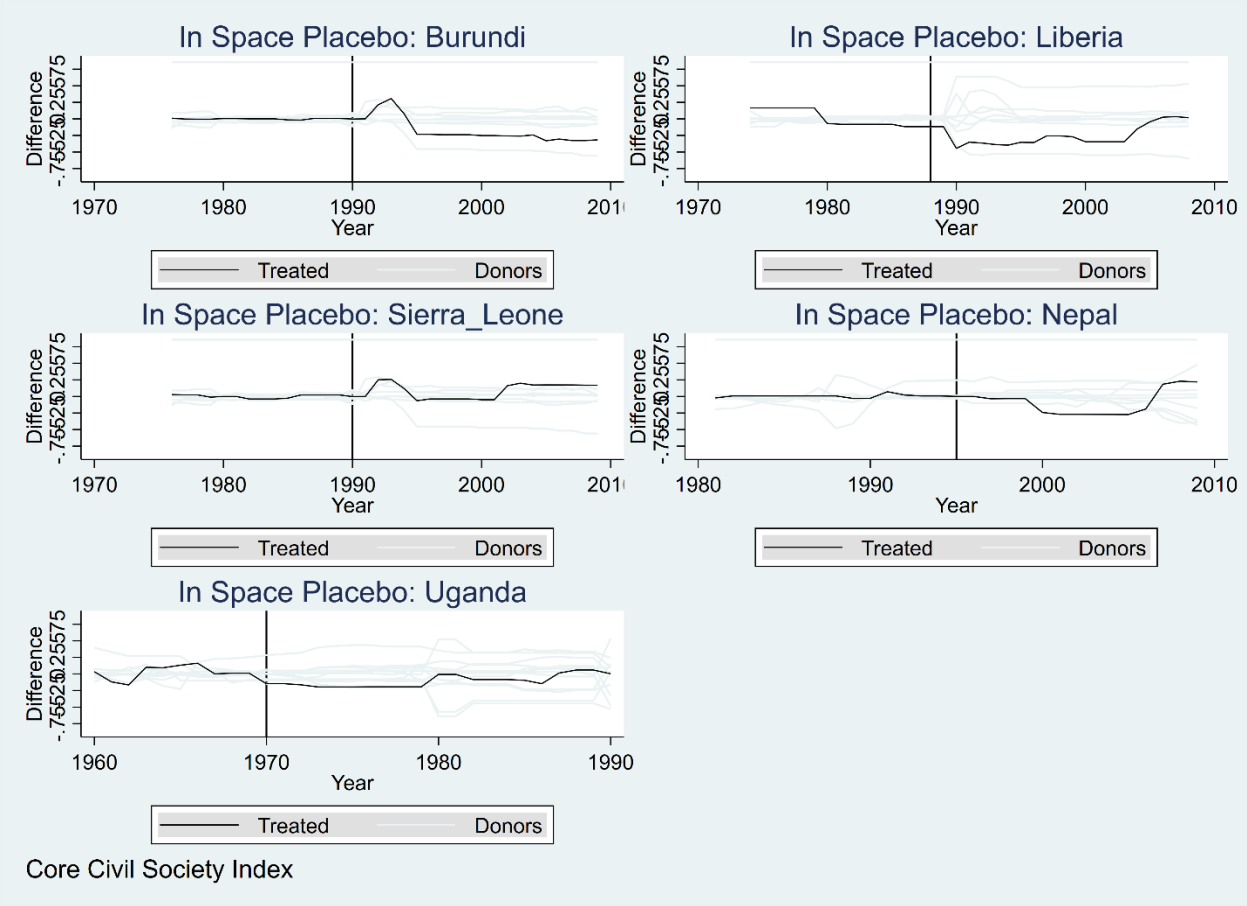
Figure A4: Synthetic Control Effects






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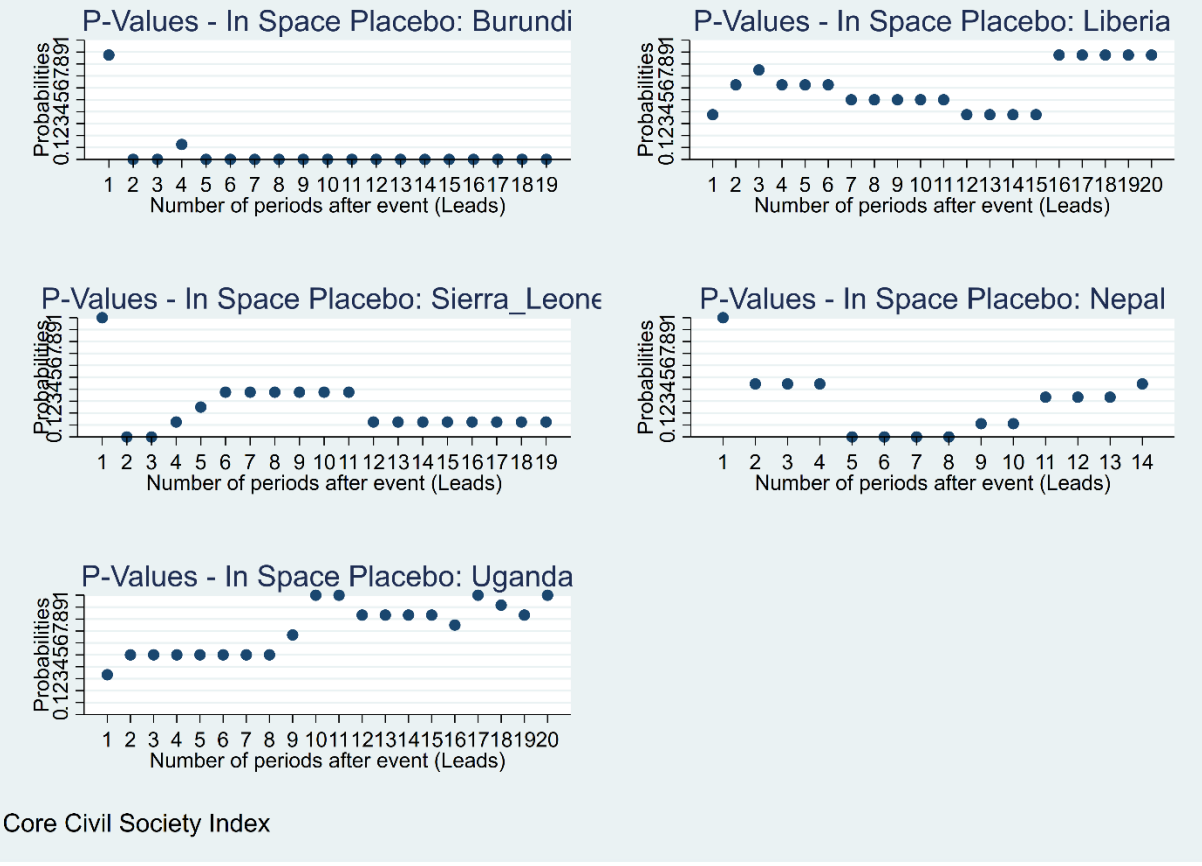
Figure A5: In Space Placebo Tests






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Figure A6: Placebo P-values






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Table A3: Fit of the Synthetic Control – Core Civil Society Index (Low Intensity)

| | Treated Covariates | Synthetic Covariates | Weights | Synthetic Country | Weight | RMSPE |
|---|---|---|---|---|---|---|
| Burundi | | | | | | |
| Civil Society (t-1) | 0.12 | 0.13 | 0.01 | Malawi | 0.81 | 0.01 |
| Civil Society (t-8) | 0.09 | 0.09 | 0.90 | Madagascar | 0.13 | |
| Civil Society (t-15) | 0.10 | 0.09 | 0.01 | Tanzania | 0.05 | |
| Log GDP per capita | 6.73 | 7.08 | 0.06 | Benin | 0.01 | |
| Log Population | 1.53 | 2.05 | 0.00 | | | |
| Electoral Democracy | 0.13 | 0.19 | 0.01 | | | |
| Distance Equator | 0.04 | 0.15 | 0.01 | | | |
| Legal Origin | 1.00 | 0.14 | 0.00 | | | |
| Human Capital | 1.11 | 1.34 | 0.00 | | | |
| Liberia | | | | | | |
| Civil Society (t-1) | 0.49 | 0.61 | 0.16 | Botswana | 0.51 | 0.13 |
| Civil Society (t-8) | 0.52 | 0.61 | 0.01 | Benin | 0.49 | |
| Civil Society (t-15) | 0.78 | 0.62 | 0.26 | | | |
| Log GDP per capita | 7.08 | 7.61 | 0.06 | | | |
| Log Population | 0.66 | 0.69 | 0.32 | | | |
| Electoral Democracy | 0.19 | 0.39 | 0.02 | | | |
| Distance Equator | 0.07 | 0.18 | 0.05 | | | |
| Legal Origin | 1.00 | 0.51 | 0.02 | | | |
| Human Capital | 1.26 | 1.33 | 0.10 | | | |
| Nepal | | | | | | |
| Civil Society (t-1) | 0.81 | 0.81 | 1.00 | Mongolia | 0.51 | 0.02 |
| Civil Society (t-8) | 0.43 | 0.43 | 0.00 | Fiji | 0.48 | |
| Civil Society (t-15) | 0.40 | 0.43 | 0.00 | Maldives | 0.01 | |
| Log GDP per capita | 6.88 | 8.13 | 0.00 | | | |
| Log Population | 2.90 | 0.20 | 0.00 | | | |
| Electoral Democracy | 0.24 | 0.44 | 0.00 | | | |
| Distance Equator | 0.32 | 0.35 | 0.00 | | | |
| Legal Origin | 1.00 | 0.49 | 0.00 | | | |
| Human Capital | 1.25 | 2.25 | 0.00 | | | |

Note: Results are obtained from estimating a synthetic control model as described in section III.



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Table A3: Fit of the Synthetic Control – Core Civil Society Index (Low Intensity) continued

| | Treated Covariates | Synthetic Covariates | Weights | Synthetic Country | Weight | RMSPE |
|---|---|---|---|---|---|---|
| Sierra Leone | | | | | | |
| Civil Society (t-1) | 0.34 | 0.34 | 0.43 | Malawi | 0.49 | 0.02 |
| Civil Society (t-8) | 0.24 | 0.28 | 0.20 | Tanzania | 0.39 | |
| Civil Society (t-15) | 0.27 | 0.24 | 0.29 | Gabon | 0.08 | |
| Log GDP per capita | 7.22 | 7.37 | 0.01 | Botswana | 0.03 | |
| Log Population | 1.21 | 2.13 | 0.00 | | | |
| Electoral Democracy | 0.20 | 0.23 | 0.01 | | | |
| Distance Equator | 0.09 | 0.11 | 0.04 | | | |
| Legal Origin | 1.00 | 0.91 | 0.02 | | | |
| Human Capital | 1.17 | 1.35 | 0.00 | | | |
| Uganda | | | | | | |
| Civil Society (t-1) | 0.24 | 0.39 | 0.00 | Tanzania | 0.81 | 0.11 |
| Civil Society (t-8) | 0.26 | 0.38 | 0.00 | Malawi | 0.19 | |
| Civil Society (t-15) | 6.76 | 7.25 | 0.00 | | | |
| Log GDP per capita | 2.11 | 2.27 | 0.00 | | | |
| Log Population | 0.25 | 0.26 | 0.01 | | | |
| Electoral Democracy | 0.02 | 0.09 | 0.00 | | | |
| Distance Equator | 1.00 | 1.00 | 0.99 | | | |
| Legal Origin | 1.15 | 1.31 | 0.00 | | | |
| Human Capital | 0.24 | 0.39 | 0.00 | | | |

Note: Results are obtained from estimating a synthetic control model as described in section III.

Table A4: F-Test for Significance

| | Core Civil Society Index | |
|---|---|---|
| | F-Statistic | P-value |
| Burundi | 5.11 | 0.001 |
| Liberia | 0.599 | 0.770 |
| Nepal | 14.672 | 0.000 |
| Sierra Leone | 3.502 | 0.008 |
| Uganda | 0.512 | 0.889 |

Note: The F-test is a joint significance test for all the post-treatment periods.



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